Sunila Maharjan

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SUMMARY

Junior Data Scientist and Software Engineer with 3+ years of software development experience and an MSc in Artificial Intelligence with Distinction. Hands-on experience using Python, SQL, Pandas, PySpark, Scikit-learn, TensorFlow, Flask and Azure for data analysis, machine learning, data pipelines, visualisation and application development. Experienced in working with large-scale healthcare datasets, developing dashboards, applying machine learning algorithms, forecasting and extracting insights from data. Strong software engineering background in RESTful APIs, databases, cloud technologies, CI/CD and Agile development, providing a strong foundation for taking data-driven solutions from development through deployment. Research experience includes deep learning, NLP, LLM technologies and computer vision-related work, including a Master's research project on sign language detection and translation using a hybrid CNN-Transformer model.

EDUCATION

London Metropolitan University
MSc Artificial Intelligence — Distinction, Artificial Intelligence, Machine Learning, Big Data, Data Analysis, Data Pipelines, Machine Learning Algorithms, Data Visualisation, AI Research

Himalaya College of Engineering, Nepal Nepal
Bachelor of Engineering (BE) Computer Engineering — First Division, 70.72%

EXPERIENCE

Research Assistant / App Developer Jan 2024 – Present
London Metropolitan University *UK*

Research Assistant – Data Science & Healthcare Analytics Jan 2024 – Jul 2024
London Metropolitan University *UK*

- Designed and implemented data pipelines and dashboards using Python, Flask and PySpark for healthcare analytics projects.
- Worked with large-scale healthcare datasets, using Azure cloud tools and data processing technologies to support analysis and visualisation.
- Applied data analysis, forecasting and visualisation techniques to identify insights from healthcare data.
- Used Python-based data processing and visualisation tools to support research findings and stakeholder reporting.
- Worked with PySpark to process and analyse large datasets.
- Used Azure cloud technologies to support data processing and visualisation workflows.
- Developed dashboards and visualisations using tools including Power BI, Plotly, Matplotlib and Seaborn.
- Led a research team working on big data and healthcare analytics, coordinating tasks and contributing to project delivery.
- Co-authored research work involving analysis of large datasets and presentation of analytical findings.
- Presented research findings to academic and industry stakeholders.
- Awarded First Place for Best Presentation at the Student & Staff Research Conference 2024.

App Developer – Easy Event Oct 2025 – Present
London Metropolitan University *UK*

- Lead a student development team in the design and delivery of a web application for a local council.
- Develop responsive applications using HTML5, JavaScript and Tailwind CSS.

- Collaborate with stakeholders to gather requirements, define user journeys and translate requirements into technical solutions.
- Coordinate project planning, task allocation and development activities within an Agile environment.
- Use Git for source control and participate in code reviews to maintain development quality.
- Conduct usability and accessibility testing and implement iterative improvements based on stakeholder feedback.
- Produce technical documentation and demonstrate application functionality to university and council stakeholders.

Software Engineer Mar 2018 – Sep 2021

Arhant Solutions Pvt. Ltd.

- Developed and maintained enterprise applications using ASP.NET (C#), JavaScript and SQL Server.
- Designed, developed and documented RESTful APIs for integration with internal and third-party systems.
- Developed database-driven applications and optimised SQL queries and database performance through query tuning and schema design.
- Worked with relational databases including SQL Server and developed strong SQL skills applicable to data analysis and data-driven applications.
- Migrated legacy desktop applications to scalable web-based platforms, improving maintainability and accessibility.
- Collaborated with cross-functional teams throughout the Software Development Life Cycle (SDLC), from requirements gathering through deployment.
- Implemented CI/CD pipelines and automated deployment processes, reducing deployment times by 60%.
- Conducted testing, debugging, code reviews and performance optimisation to support reliable software delivery.
- Maintained source code using Git and SVN.
- Implemented secure authentication and authorisation using JWT and role-based access controls.
- Contributed to system architecture discussions and followed software development and security best practices.

Bootcamp Participant Jan 2025 – Mar 2025

Code Cumbria | Scale-Ability

- Completed an intensive 10-week technical bootcamp covering Python, SQL, Data Science, Machine Learning, Data Visualisation, Digital Logic, PLC programming, cyber security, SQL, microcontrollers and Arduino.
- Developed hands-on skills in Python, SQL, data science and machine learning through practical projects.
- Applied data visualisation, reporting and automation techniques to real-world business problems.
- Strengthened collaborative software development and source control practices using GitHub.
- Developed practical understanding of machine learning workflows, data analysis and software development practices.

PROJECTS

MSc Research Project – Sign Language Detection and Translation Using Smart Glove | London Metropolitan University

- Developed a research project focused on sign language detection and translation using sensor data collected from a smart glove.
- Applied machine learning and deep learning techniques to recognise sign language patterns.
- Research work included development and evaluation of a hybrid CNN-Transformer model.
- Explored the application of AI techniques to real-world assistive technology problems.
- Project was selected by the university's Assistive Technology Research Group.
- Related research publications: Sign Language Detection and Translation Using Smart Glove.
- Related research publications: Sign Language Detection and Translation Using Smart Glove 2.0 Using Hybrid CNN-Transformer Model.

Big Data & Healthcare Analytics

- Worked with large-scale healthcare datasets using Python, PySpark and Azure.
- Developed data pipelines to support data processing and analysis.
- Created dashboards and visualisations to communicate analytical findings.

- Applied forecasting and data analysis techniques to healthcare-related datasets.
- Presented analytical findings to academic and industry stakeholders.

TECHNICAL SKILLS

Programming & Data: Python, SQL, JavaScript, C#, Pandas, NumPy, PySpark, Data Manipulation, Data Analysis

Machine Learning & AI: Machine Learning, Scikit-learn, TensorFlow, NLP, LLM, Deep Learning, Predictive Analytics, Forecasting, Statistical Analysis, Computer Vision

Data Engineering & Big Data: Data Pipelines, PySpark, Hadoop, Large-Scale Data Processing, Data Visualisation, Data Analysis

Databases: SQL, PostgreSQL, SQLAlchemy, SQL Server, Oracle, SQLite

Data Visualisation & Reporting: Power BI, Pandas, Plotly, Matplotlib, Seaborn, Dashboards, Data Visualisation

Cloud, APIs & Deployment: Microsoft Azure, Docker, Flask, RESTful APIs, Web Services, CI/CD Pipelines, GitHub Actions

Software Development: ASP.NET (C#), JavaScript, HTML5, Responsive Web Development, API Development, Authentication & Authorisation, Testing, Debugging

Tools & Version Control: Git, GitHub, SVN, Jupyter Notebook, Visual Studio, VS Code

Methodologies: Agile Development, SDLC, Code Reviews, Technical Documentation, Project Management, Team Leadership

Additional Skills: Project Management, Team Leadership, Teamwork, Communication, Critical Thinking, Problem Solving, Time Management, Adaptability, Technical Documentation, Stakeholder Collaboration

RESEARCH PUBLICATIONS

Sign Language Detection and Translation Using Smart Glove — Innovative Computing and Communications (ICICC-2024) Vitamin D Analysis for Sustainable Healthcare in Inner London Borough — Computational Intelligence Sign Language Detection and Translation Using Smart Glove 2.0 Using Hybrid CNN-Transformer Model — Proceedings of Data Analytics and Management (ICDAM-2025)